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United States Patent Application Publication

20250279327

Kind Code

A1

Publication Date

September 04, 2025

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SEMICONDUCTOR DEVICE

Abstract

Provided is a semiconductor device including a semiconductor substrate, where the semiconductor substrate has: a temperature sensing diode provided above an upper surface of the semiconductor substrate; a diode wiring provided above the upper surface of the semiconductor substrate and connected to the temperature sensing diode; a metal layer provided in at least part of a region between the diode wiring and the upper surface of the semiconductor substrate; a first interlayer dielectric film provided between the metal layer and the upper surface of the semiconductor substrate; and a second interlayer dielectric film provided between the diode wiring and the metal layer.

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Family ID: 96880484

Appl. No.: 19/037267

Filed: January 26, 2025

Foreign Application Priority Data

JP 2024-030805

Feb. 29, 2024

Publication Classification

Int. Cl.: H01L23/34 (20060101); H10D12/00 (20250101); H10D64/00 (20250101)

U.S. Cl.:

CPC H01L23/34 (20130101); H10D12/481 (20250101); H10D64/117 (20250101);

Background/Summary

The contents of the following patent application(s) are incorporated herein by reference: NO. 2024-030805 filed in JP on Feb. 29, 2024.

BACKGROUND

1. Technical Field

[0001] The present invention relates to a semiconductor device.

2. Related Art

[0002] In the related art, a semiconductor device has been known in which a thermosensitive element can be disposed in a state where the element is electrically separated from a semiconductor substrate without being limited by the diffusing structure of a semiconductor layer underlying the thermosensitive element (see Patent Document 1, for example). In addition, a semiconductor device has been known which comprises a temperature sense part having a stable temperature detection accuracy while suppressing increase in chip area (See Patent Document 2, for example).

Prior Art Documents

Patent Documents

[0003] Patent Document 1: Japanese Patent Application Publication No. **2005-26279** [0004] Patent Document 2: Japanese Patent No. **6872933**

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1A** illustrates a top view of one example of a semiconductor device **100** according to one embodiment of the present invention.

[0006] FIG. **1B** illustrates arrangement of a temperature sensing diode **66**, a diode wiring **60**, and a metal layer **80** in a top view.

[0007] FIG. **2** illustrates one example of the semiconductor device **100** in a cross section A-A' shown in FIG. **1B**.

[0008] FIG. **3** illustrates a cross section A-A' of a semiconductor device **200** according to a comparative example.

[0009] FIG. **4** illustrates one example of the semiconductor device **100** in a cross section B-B' shown in FIG. **1B**.

[0010] FIG. **5** illustrates one example of the semiconductor device **100** in a cross section C-C' shown in FIG. **1B**.

[0011] FIG. **6** illustrates one example of a top view of the semiconductor device **100**.

[0012] FIG. **7** illustrates one example of the semiconductor device **100** in a cross section D-D' shown in FIG. **6**.

[0013] FIG. **8** illustrates a top view of one example of a semiconductor device **300** according to another embodiment of the present invention.

[0014] FIG. **9** illustrates one example of the semiconductor device **300** in a cross section D-D' shown in FIG. **8**.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0015] Hereinafter, the present invention will be described through embodiments of the invention, but the following embodiments do not limit the invention according to claims. In addition, not all combinations of features described in the embodiments are essential to a solution of the invention. It should be noted that, for elements having substantially a same function and configuration in the present specification and drawings, a same reference numeral will be given to the elements to avoid redundant description thereof, and elements which are not directly related to the present invention

will not be shown. In addition, for elements having substantially a same function and configuration in one drawing, a reference numeral may be given to an element as a representative, and for the rest of the elements, the reference numeral may not be given.

[0016] In the present specification, one side in a direction parallel to a depth direction of a semiconductor substrate is referred to as “upper” and another side is referred to as “lower”. One surface of two principal surfaces of a substrate, a layer, or another member is referred to as an upper surface, and another surface is referred to as a lower surface. “Upper” and “lower” directions are not limited to a direction of gravity, or a direction in which a semiconductor module is mounted.

[0017] In the present specification, technical matters may be described using orthogonal coordinate axes of an X axis, a Y axis, and a Z axis. The orthogonal coordinate axes merely specify relative positions of components, and do not limit a specific direction. For example, a Z axis is not limited to indicate the height direction with respect to the ground. It should be noted that a +Z axis direction and a -Z axis direction are directions opposite to each other. If a Z axis direction is described without describing the signs, it means that the direction is parallel to a +Z axis and a -Z axis. In the present specification, orthogonal axes parallel to the upper surface and the lower surface of the semiconductor substrate are referred to as the X axis and the Y axis. In addition, an axis perpendicular to the upper surface and the lower surface of the semiconductor substrate is referred to as the Z axis. In the present specification, the direction of the Z axis may be referred to as the depth direction. In addition, in the present specification, a direction parallel to the upper surface and the lower surface of the semiconductor substrate may be referred to as a horizontal direction, including an X axis direction and a Y axis direction.

[0018] A case where a term such as “same” or “equal” is used in the present specification may include a case where an error due to a variation in manufacturing or the like is included. The error is, for example, within 10%.

[0019] FIG. 1A illustrates a top view of one example of a semiconductor device **100** according to one embodiment of the present invention. FIG. 1A shows a position obtained by projecting a position of each member onto an upper surface of a semiconductor substrate **10**. FIG. 1A shows only some members of the semiconductor device **100**, and does not show some members.

[0020] The semiconductor device **100** includes the semiconductor substrate **10**. The semiconductor substrate **10** is a substrate made of a compound semiconductor such as silicon or SiC. The semiconductor substrate **10** includes an end side **102** in a top view. The semiconductor substrate **10** in the present example includes two sets of end sides **102** opposite each other in a top view. FIG. 1A shows one set of end sides **102-1** and **102-2** opposite each other. In FIG. 1A, a direction parallel to the end side **102-1** and the end side **102-2** is referred to as a Y axis direction, and a direction perpendicular to the end side **102-1** and the end side **102-2** is referred to as an X axis direction.

[0021] The semiconductor substrate **10** is provided with an active portion **120**. The semiconductor substrate **10** in the present example is provided with an active portion **120-1** and an active portion **120-2**. The active portion **120** is a region through which a main current flows in a depth direction between the upper surface and a lower surface of the semiconductor substrate **10** if the semiconductor device **100** is controlled to be in an on-state.

[0022] The active portion **120** is provided with a semiconductor element. The semiconductor element may be a transistor element such as an IGBT, may be a diode element such as an FWD, or may be both of them.

[0023] The semiconductor device **100** includes an upper surface electrode **52** provided above the upper surface of the semiconductor substrate **10**. The upper surface electrode **52** is provided above the active portion **120**. The upper surface electrode **52** may be provided above each active portion **120**. The semiconductor device **100** in the present example includes an upper surface electrode **52-1** and an upper surface electrode **52-2**. The upper surface electrode **52-1** is provided above the active portion **120-1**, and the upper surface electrode **52-2** is provided above the active portion **120-**

2. In the present specification, the upper surface electrode **52-1** and the upper surface electrode **52-2** may respectively be referred to as a first portion and a second portion. The upper surface electrode **52** is, for example, an Al-Si alloy.

[0024] The upper surface electrode **52-1** and the upper surface electrode **52-2** may be electrically connected to each other via at least one of a metal layer, solder, or a lead frame described later. The upper surface electrode **52** may be an electrode having largest area in a top view among electrodes provided above the upper surface of the semiconductor substrate **10**. The upper surface electrode **52** may be electrically connected to an emitter region or a source region of a transistor portion, or may be electrically connected to an anode region of a diode portion. The active portion **120** may be a region overlapping with the upper surface electrode **52** in a top view.

[0025] The semiconductor substrate **10** is provided with a well region of a P type. The well region is a P type region having a higher concentration than a base region of the transistor portion or the anode region of the diode portion. The base region is a P type region: provided facing a conductive portion of a gate electrode; and having a channel formed in a portion facing the conductive portion if a predetermined gate voltage is applied to the conductive portion. The semiconductor substrate **10** includes a first well region **111** and a second well region **112**. The first well region **111** and the second well region **112** are provided sandwiching the active portion **120** in a top view. The first well region **111** and the second well region **112** are provided sandwiching the active portion **120** in a predetermined direction (the X axis direction in FIG. 1A). The two well regions sandwiching the active portion **120** means that any straight line connecting the two well regions in a top view passes through the active portion **120**. The first well region **111** and the second well region **112** in the present example are both rectangular shaped. In FIG. 1A, the first well region **111** and the second well region **112** are hatched.

[0026] The first well region **111** may be provided near the end side **102-1**. That is, a distance between the first well region **111** and the end side **102-1** is smaller than a distance between the first well region **111** and the end side **102-2**. The second well region **112** may be provided near the end side **102-2**. That is, a distance between the second well region **112** and the end side **102-2** is smaller than a distance between the second well region **112** and the end side **102-1**.

[0027] The first well region **111** in the present example is provided between the active portion **120** and the end side **102-1** in the X axis direction. The active portion **120** is not provided between the first well region **111** and the end side **102-1**. That is, the first well region **111** is provided between an end portion of the active portion **120** in the X axis direction and the end side **102-1**.

[0028] The second well region **112** in the present example is provided between the active portion **120** and the end side **102-2** in the X axis direction. The active portion **120** is not provided between the second well region **112** and the end side **102-2**. That is, the second well region **112** is provided between an end portion of the active portion **120** in the X axis direction and the end side **102-2**.

[0029] The first well region **111** and the second well region **112** may be provided in a range including a central position X_c of the end side **102-1** and the end side **102-2** in the Y axis direction. The first well region **111** may be sandwiched between active portions **120** in the Y axis direction. The second well region **112** may be sandwiched between the active portions **120** in the Y axis direction. The second well region **112** may be provided in a range which is wider than the first well region **111** in the Y axis direction.

[0030] The semiconductor substrate **10** may include a peripheral well region **113** provided enclosing the active portion **120** in a top view. The peripheral well region **113** may be provided parallel to each end side of the semiconductor substrate **10**. The peripheral well region **113** is in contact with each end side of the semiconductor substrate **10** in FIG. 1A, but the peripheral well region **113** may be away from each end side. The peripheral well region **113** in the present example is an annular region enclosing the active portion **120** in a top view. The peripheral well region **113** may have a constant width in a direction perpendicular to each end side. In FIG. 1A, the peripheral well region **113** is also hatched. It is to be noted that portions, which overlap with a gate pad **50**, a

current detection pad **72**, an anode pad **74**, and a cathode pad **76**, of respective well regions are not hatched.

[0031] The first well region **111** and the second well region **112** in the present example protrude farther toward a center of the active portion **120** than the peripheral well region **113**. In another example, at least one of the first well region **111** or the second well region **112** may be provided between the peripheral well region **113** and the end side **102** of the semiconductor substrate **10**. In this case, the first well region **111** and the second well region **112** protrude from the peripheral well region **113** toward the end side **102**.

[0032] The semiconductor substrate **10** may include a dividing well region **114** which divides the active portion **120** in a top view. The active portion **120** may be divided into the active portion **120-1** and the active portion **120-2** by a well region including the dividing well region **114**. The dividing well region **114** is sandwiched between the active portions **120** in a top view. The dividing well region **114** has a longitudinal part in a predetermined well longitudinal direction. The dividing well region **114** extends in the well longitudinal direction to cross the active portion **120**. The well longitudinal direction of the dividing well region **114** is the X axis direction. In FIG. **1A**, the dividing well region **114** is also hatched.

[0033] The dividing well region **114** may be provided between the first well region **111** and the second well region **112**. One end and another end, in a longitudinal direction, of the dividing well region **114** may respectively be connected to the first well region **111** and the second well region **112**. The dividing well region **114** may be provided in a region overlapping with the center of the active portion **120**.

[0034] The dividing well region **114** may include a wide portion **115** having a width larger than that of another portion in a direction perpendicular to the well longitudinal direction in a top view (the Y axis direction in the present example). The wide portion **115** is also provided between the first well region **111** and the second well region **112**. The wide portion **115** may be provided in a region overlapping with the center of the active portion **120**. The wide portion **115** may be provided in a region including a center of the dividing well region **114** in the well longitudinal direction.

[0035] The semiconductor substrate **10** in the present example includes control electrodes such as the gate pad **50**, the current detection pad **72**, the anode pad **74**, and the cathode pad **76**. The gate pad **50** is provided above the first well region **111**. The current detection pad **72**, the anode pad **74**, and the cathode pad **76** are provided above the second well region **112**.

[0036] A predetermined gate voltage is applied to the gate pad **50**. The gate voltage applied to the gate pad **50** is supplied to a transistor portion of the active portion **120** by a gate runner **48** described later or the like.

[0037] The semiconductor substrate **10** includes the gate runner **48** provided above the upper surface of the semiconductor substrate **10**. In FIG. **1A**, the gate runner **48** is indicated by a dashed line. In the present example, the gate runner **48** is a wiring made of polysilicon to which an impurity is added. The gate runner **48** may be made of a conductive material such as a metal. The gate runner **48** supplies the gate voltage applied to the gate pad **50** to the transistor portion provided in the active portion **120**.

[0038] The gate runner **48** may be provided above each well region. Part of the gate runner **48** may be provided along the peripheral well region **113** so as to enclose the active portion **120**. In addition, another gate runner **48** may also be provided between each control electrode and the active portion **120** along the first well region **111** and the second well region **112**. Another gate runner **48** may be provided above the dividing well region **114**, and may overlap with a temperature sensing diode **66** and a diode wiring **60** described later. That is, the gate runner **48** in the present example may divide the active portion **120** in a top view.

[0039] The current detection pad **72** is connected to a current detection unit (not shown), and detects a current flowing through the current detection unit.

[0040] FIG. **1B** illustrates arrangement of the temperature sensing diode **66**, the diode wiring **60**,

and a metal layer **80** in a top view. In FIG. **1B**, the well regions are not hatched, and the temperature sensing diode **66** and the metal layer **80** are hatched.

[0041] The semiconductor substrate **10** includes the temperature sensing diode **66** provided above the upper surface of the semiconductor substrate **10**. The temperature sensing diode **66** detects temperature of the semiconductor substrate **10**. For example, the temperature sensing diode **66** is a PN junction diode including an anode region and a cathode region made of a semiconductor material such as polysilicon.

[0042] The semiconductor substrate **10** includes the diode wiring **60**. The diode wiring **60** is provided above the upper surface of the semiconductor substrate **10**, and is connected to the temperature sensing diode **66**. The diode wiring **60** includes an anode wiring **60-1** and a cathode wiring **60-2**. The diode wiring **60** may be made of a semiconductor material such as polysilicon, or may be made of a metal.

[0043] The anode pad **74** is connected to the anode region of the temperature sensing diode **66** via the anode wiring **60-1** of the diode wiring **60**. The cathode pad **76** is connected to the cathode region of the temperature sensing diode **66** via the cathode wiring **60-2** of the diode wiring **60**. For example, the temperature of the semiconductor substrate **10** is measured by measuring forward voltages of the temperature sensing diode **66** at the anode pad **74** and the cathode pad **76**. The diode wiring **60**, the temperature sensing diode **66**, and the gate runner **48** in the present example overlap with the dividing well region **114** in a top view.

[0044] The upper surface electrode **52** and the control electrodes each contain a metal such as aluminum. An interlayer dielectric film is provided between the upper surface electrode **52** and each control electrode, and the semiconductor substrate **10**. The upper surface electrode **52** and the semiconductor substrate **10** are connected to each other via a contact hole provided in the interlayer dielectric film. FIG. **1B** does not show the interlayer dielectric film and the contact hole.

[0045] The upper surface electrode **52** in the present example does not overlap with the diode wiring **60** in a top view. The upper surface electrode **52** in the present example does not overlap with the temperature sensing diode **66** in a top view, either. The diode wiring **60** and the temperature sensing diode **66** in the present example are sandwiched between the first portion **52-1** and the second portion **52-2** in a top view. The diode wiring **60** and the temperature sensing diode **66** in the present example are sandwiched between the active portion **120-1** and the active portion **120-2** in a top view.

[0046] The semiconductor substrate **10** includes the metal layer **80**. The metal layer **80** is provided in at least part of a region between the diode wiring **60** and the upper surface of the semiconductor substrate **10** (see FIG. **2**). FIG. **1B** shows only the metal layer **80** provided near the diode wiring **60**. It is to be noted that the metal layer **80** may also be provided in a region other than a range overlapping with the diode wiring **60** in a top view. For example, the metal layer **80** is also provided in the active portion **120**.

[0047] The semiconductor substrate **10** may include an edge termination structure portion between the peripheral well region **113** and the end side **102** of the semiconductor substrate **10**. The edge termination structure portion reduces electric field strength in the semiconductor substrate **10** on a side of the upper surface thereof. The edge termination structure portion has, for example, a guard ring annularly provided enclosing the active portion **120**, a field plate, a RESURF, and a structure combining them. The present specification does not illustrate the edge termination structure portion.

[0048] FIG. **2** illustrates one example of the semiconductor device **100** in a cross section A-A' shown in FIG. **1B**. The cross section A-A' is a YZ cross section perpendicular to a direction in which the diode wiring **60** extends (the X axis direction). In the cross section, the semiconductor device **100** includes the semiconductor substrate **10**, solder **35**, plating **36**, an interlayer dielectric film **38**, the gate runner **48**, the upper surface electrode **52**, the diode wiring **60**, the metal layer **80**, a lead frame **90**, and a wiring protection film **92**. In the cross section, a lower surface of the

semiconductor substrate **10** is not shown.

[0049] The semiconductor substrate **10** is provided with a trench portion **40**. The trench portion **40** is provided extending from an upper surface **21** to an inside of the semiconductor substrate **10**. The trench portion **40** may be a gate trench. The trench portion **40** includes a dielectric film **42** and a conductive portion **44** therein. The conductive portion **44** may be electrically connected to the gate pad **50** via the gate runner **48**. In FIG. 2, the conductive portion **44** is hatched. The dielectric film **42** insulates the semiconductor substrate **10** and the conductive portion from each other. It is to be noted that the trench portion **40** may be a dummy trench. In that case, the conductive portion **44** may be electrically connected to the upper surface electrode **52**.

[0050] In the cross section, the dividing well region **114** of a P+ type is provided within the semiconductor substrate **10**. The dividing well region **114** is provided extending from the upper surface **21** of the semiconductor substrate **10** to a depth greater than that of a lower end of the trench portion **40**. An end portion, in an extending direction, of the trench portion **40** (the Y axis direction in the present example) may be located within the dividing well region **114**. In the cross section, a drift region **18** of an N-type is provided below the trench portion **40** and the dividing well region **114**. In addition, in FIG. 2, in order to show a size of the dividing well region **114**, the trench portion **40** extending into the dividing well region **114** is not shown, but the dividing well region **114** and the conductive portion **44** are electrically insulated from each other by the dielectric film **42**.

[0051] The semiconductor substrate **10** includes a first interlayer dielectric film **38-1** provided on the upper surface **21**. The first interlayer dielectric film **38-1** is provided between the metal layer **80** and the upper surface **21** of the semiconductor substrate **10**. The first interlayer dielectric film **38-1** may be in contact with the upper surface **21**. The first interlayer dielectric film **38-1** insulates the metal layer **80** and the upper surface electrode **52**, and the semiconductor substrate **10** from each other. In a case of the present example, the first interlayer dielectric film **38-1** also insulates the metal layer **80** and the gate runner **48** from each other. The interlayer dielectric film **38** is provided between the gate runner **48** and the upper surface **21** of the semiconductor substrate **10**. The interlayer dielectric film **38** may be a dielectric film different from the first interlayer dielectric film **38-1**.

[0052] The semiconductor substrate **10** includes the metal layer **80** provided above the upper surface **21**. The metal layer **80** is provided in at least part of a region between the diode wiring **60** and the upper surface **21** of the semiconductor substrate **10**. The metal layer **80** may be provided over an entire horizontal direction between the diode wiring **60** and the upper surface **21** of the semiconductor substrate **10**, at a position overlapping with the diode wiring **60** in the depth direction. The metal layer **80** in the present example is provided not only below the diode wiring **60** but also in the active portion **120**. The metal layer **80** may also be provided below the plating **36**.

[0053] The semiconductor substrate **10** includes a second interlayer dielectric film **38-2** provided above the upper surface **21**. The second interlayer dielectric film **38-2** is provided between the diode wiring **60** and the metal layer **80**, and insulates the diode wiring **60** and the metal layer **80** from each other. The first interlayer dielectric film **38-1** and the second interlayer dielectric film **38-2** each include at least one of a dielectric film such as a silicate glass to which an impurity such as boron or phosphorus is added, a thermal oxide film, or another dielectric film.

[0054] The diode wiring **60** is provided above the second interlayer dielectric film **38-2**. As described above, the diode wiring **60** includes the anode wiring **60-1** and the cathode wiring **60-2**. The diode wiring **60** in the present example is covered with the wiring protection film **92**. The wiring protection film **92** is, for example, polyimide.

[0055] The upper surface electrode **52** is provided above the metal layer **80**. The upper surface electrode **52** is connected to the metal layer **80**. The upper surface electrode **52** and the metal layer **80** may be mutually laminated. As described above, the upper surface electrode **52** includes the first portion **52-1** and the second portion **52-2**. In a top view, the first portion **52-1** and the second

portion **52-2** are arranged with the diode wiring **60** therebetween.

[0056] The plating **36** is provided above the upper surface electrode **52**. The upper surface electrode **52** is connected to the lead frame **90**, which is a wiring member, via the plating **36**. The plating **36** is, for example, Ni plating. Providing the plating **36** can improve wettability of the upper surface electrode **52** and the solder **35**, which is a bonding member, and increase bondability of the lead frame **90**. The plating **36** may be provided over an entirety of an upper surface of the upper surface electrode **52**, or may be provided over most of the upper surface of the upper surface electrode **52** (for example, larger than or equal to half of area of the upper surface).

[0057] The solder **35** is provided above the plating **36**. In the present example, the solder **35** is provided between the lead frame **90** and the plating **36**. In addition, in FIG. 2, the solder **35** is also provided above the wiring protection film **92**. In the present example, the solder **35** is provided between the lead frame **90** and the wiring protection film **92**. It is to be noted that the solder **35** may not be provided above the wiring protection film **92**.

[0058] The lead frame **90** is provided above the solder **35**. The upper surface electrode **52** is connected to an external circuit via the lead frame **90**.

[0059] At least part of the solder **35** may be provided at a same height as that of the wiring protection film **92**. In addition, the wiring protection film **92** is provided at a position lower than the lead frame **90** in a height direction (a Z axis direction). In the present example, the wiring protection film **92** is provided below the lead frame **90**.

[0060] In the present example, there is a point where the solder **35**, the plating **36**, and the wiring protection film **92** come into contact. The point is referred to as a triple point **G1**. With presence of the triple point **G1**, stress repeatedly applied due to heat may cause stress concentration at the triple point **G1**, forming a crack from the triple point toward the upper surface **21** of the semiconductor substrate **10**. In a case of the present example, the metal layer **80** can suppress development of the crack.

[0061] FIG. 3 illustrates a cross section A-A' of a semiconductor device **200** according to a comparative example. In the semiconductor device **200**, the metal layer **80** is not provided between the diode wiring **60** and the upper surface **21** of the semiconductor substrate **10**. Accordingly, the second interlayer dielectric film **38-2** is not provided, either. Except for that, the semiconductor device **200** has a configuration similar to that of the semiconductor device **100**.

[0062] In the semiconductor device **200**, between the diode wiring **60** and the upper surface **21** of the semiconductor substrate **10**, the metal layer **80** is not provided, and only the first interlayer dielectric film **38-1** with low strength and the gate runner **48** made of polysilicon are provided. Therefore, if the crack from the triple point **G1** described above develops between the diode wiring **60** and the upper surface **21** of the semiconductor substrate **10**, the development of the crack may not be suppressed. This may result in, for example, a problem such as development of the crack in the first interlayer dielectric film **38-1**, causing continuity between the diode wiring **60** and the gate runner **48**.

[0063] On the other hand, in the semiconductor device **100** according to an embodiment, the metal layer **80** provided between the diode wiring **60** and the upper surface **21** of the semiconductor substrate **10** can improve strength of a part below the diode wiring **60**. This makes it possible to suppress the development of the crack to the part below the diode wiring **60**. Further, providing the metal layer **80** between the diode wiring **60** and the upper surface **21** of the semiconductor substrate **10** can protect the diode wiring **60** from noise in the semiconductor substrate **10**. In addition, the second interlayer dielectric film **38-2** can also improve the strength of the part below the diode wiring **60**, and protect the diode wiring **60** from the noise.

[0064] The metal layer **80** may be composed of an electrically conductive material. This increases an effect of protection against noise. In addition, from a viewpoint of improving the strength, it is preferable that hardness of the metal layer **80** is high. For example, the metal layer **80** contains a metal harder than aluminum. The metal layer **80** and the upper surface electrode **52** may be made

of different materials. The metal layer **80** may contain a harder metal than the upper surface electrode **52**. The metal layer **80** may contain at least one of tungsten, titanium, or tantalum. This can further improve the strength of the part below the diode wiring **60**.

[0065] In FIG. **2**, side surfaces of the second interlayer dielectric film **38-2** may be in contact with the upper surface electrode **52**. One of the side surfaces of the second interlayer dielectric film **38-2** in the present example is in contact with the first portion **52-1**, and another is in contact with the second portion **52-2**. In other words, the second interlayer dielectric film **38-2** is sandwiched between upper surface electrodes **52** in a direction (the Y axis direction) perpendicular to the direction in which the diode wiring **60** extends (the X axis direction), in a plane parallel to the upper surface **21**.

[0066] The metal layer **80** in the present example is provided extending from the first portion **52-1**, passing below the diode wiring **60**, to the second portion **52-2**. This makes it possible to cover an entire surface of the part below the diode wiring **60**, so that the semiconductor device **100** more reliably achieves improvement of the strength and the effect of protection against noise.

[0067] The diode wiring **60** in the present example overlaps with the dividing well region **114** in the depth direction. The first interlayer dielectric film **38-1** in the present example is provided with a contact hole **45** which electrically connects the metal layer **80** and the dividing well region **114** to each other. The upper surface electrode **52** is electrically connected to the dividing well region **114** via the contact hole **45**. This makes a potential of the dividing well region **114** equal to a potential of the upper surface electrode **52**, so that the diode wiring **60** and the gate runner **48** provided above the dividing well region **114** can be protected from the noise in the semiconductor substrate **10**.

[0068] A connection portion **46** may be provided below the contact hole **45**. The connection portion **46** may be made of a same material as that of the gate runner **48**. That is, the connection portion **46** may be a wiring made of polysilicon to which an impurity is added. Providing the connection portion **46** makes it easier to electrically connect the dividing well region **114** and the upper surface electrode **52** to each other.

[0069] The first interlayer dielectric film **38-1** in the present example is provided with two contact holes **45**. The gate runner **48** in the present example is provided between the two contact holes **45** in a top view.

[0070] The dividing well region **114** in the present example is sandwiched between the active portions **120** in the Y axis direction. That is, the dividing well region **114** crosses the active portion **120**. End portions of the active portions **120** in the Y axis direction shown in FIG. **2** may match end portions of the dividing well region **114**.

[0071] In the cross section A-A', a width T1 of the second interlayer dielectric film **38-2** in a horizontal direction parallel to the upper surface **21** of the semiconductor substrate **10** may be smaller than a width T2 of the dividing well region **114**. In the cross section A-A', an entirety of the second interlayer dielectric film **38-2** may overlap with the dividing well region **114**.

[0072] FIG. **4** illustrates one example of the semiconductor device **100** in a cross section B-B' shown in FIG. **1B**. The cross section B-B' is a YZ cross section perpendicular to the direction in which the diode wiring **60** extends (the X axis direction). A configuration similar to that of the cross section A-A' shown in FIG. **2** will not be described as appropriate.

[0073] The cross section is a cross section crossing the temperature sensing diode **66**. The temperature sensing diode **66** is provided above the upper surface **21** of the semiconductor substrate **10**. A position of the temperature sensing diode **66** in the Z axis direction may be the same as a position of the diode wiring **60**. The temperature sensing diode **66** in the present example is covered with the wiring protection film **92**. The temperature sensing diode **66** in the present example overlaps with the gate runner **48** in the Z axis direction.

[0074] The metal layer **80** in the present example is also provided in at least part of a region between the temperature sensing diode **66** and the upper surface **21** of the semiconductor substrate

10. Arrangement of the first interlayer dielectric film **38-1** and the second interlayer dielectric film **38-2** is similar to that in FIG. 2. The second interlayer dielectric film **38-2** in the present example insulates the metal layer **80** and the temperature sensing diode **66** from each other. This makes it possible to improve strength of a part below the temperature sensing diode **66** similarly to a case of the diode wiring **60**. In addition, this makes it possible to protect the temperature sensing diode **66** from noise. The metal layer **80** may be provided over an entire horizontal direction between the temperature sensing diode **66** and the upper surface **21** of the semiconductor substrate **10**, at a position overlapping with the temperature sensing diode **66** in the depth direction. The metal layer **80** provided below the temperature sensing diode **66** may also be continuous with the metal layer **80** provided in the active portion **120**.

[0075] FIG. 5 illustrates one example of the semiconductor device **100** in a cross section C-C' shown in FIG. 1B. The cross section C-C' is an XZ cross section crossing the active portion **120**. In the cross section, the semiconductor substrate **10** is provided with an IGBT. It is to be noted that the semiconductor substrate **10** may be provided with a MOSFET, or may be provided with a diode.

[0076] In the cross section C-C', the semiconductor device **100** includes the semiconductor substrate **10**, the first interlayer dielectric film **38-1**, the metal layer **80**, the upper surface electrode **52**, and a lower surface electrode **24**. The first interlayer dielectric film **38-1** is provided on the upper surface **21** of the semiconductor substrate **10**. The first interlayer dielectric film **38-1** may be the same as the first interlayer dielectric film **38-1** described with reference to FIG. 2 or the like. That is, the first interlayer dielectric film **38-1** may be provided extending from below the diode wiring **60** to the active portion **120**.

[0077] The upper surface electrode **52** is provided above the first interlayer dielectric film **38-1**. In the cross section C-C', the second portion **52-2** of the upper surface electrode **52** is provided. The upper surface electrode **52** is in contact with part of the upper surface **21** of the semiconductor substrate **10** through a contact hole **54** formed in the first interlayer dielectric film **38-1**. The upper surface electrode **52** may also be considered to be in contact with the upper surface **21** of the semiconductor substrate **10** if it is in contact with the upper surface **21** via the metal layer **80**. A tungsten plug or the like for contact may be provided between the upper surface **21** of the semiconductor substrate **10** and the upper surface electrode **52**. The upper surface electrode **52** may also be considered to be in contact with the upper surface **21** of the semiconductor substrate **10** if it is in contact with the upper surface **21** via the tungsten plug.

[0078] The metal layer **80** is provided between the first interlayer dielectric film **38-1** and the upper surface electrode **52**. The metal layer **80** in the present example is electrically connected to the upper surface electrode **52** in the cross section C-C'.

[0079] The metal layer **80** may be in contact with the upper surface **21** of the semiconductor substrate **10** in a region which does not overlap with the diode wiring **60** and the temperature sensing diode **66**. The metal layer **80** in the present example is in contact with the upper surface **21** of the semiconductor substrate **10** via the contact hole **54** in the first interlayer dielectric film **38-1** in the cross section C-C'. The metal layer **80** may be in contact with a plurality of mesa portions **70**. The metal layer **80** may be provided above a plurality of trench portions.

[0080] The metal layer **80** may function as a barrier metal in the active portion **120**. Providing the metal layer **80** in the active portion **120** can prevent a resin ion from a protective film or the like provided above the upper surface electrode **52** from penetrating into the semiconductor substrate **10**. The metal layer **80** may be provided extending from below the diode wiring **60** to the active portion **120**. The metal layer **80** may have a same composition as that of the metal layer **80** below the diode wiring **60**. This allows simplification of a manufacturing process. The metal layer **80** may have a same film thickness as that of the metal layer **80** below the diode wiring **60**. This improves in-plane uniformity.

[0081] The lower surface electrode **24** is provided on the lower surface **23** of the semiconductor substrate **10**. The upper surface electrode **52** and the lower surface electrode **24** are each made of a

metal material such as aluminum.

[0082] The semiconductor substrate **10** includes the drift region **18** of the N-type. The drift region **18** may be a region, which is left without a dopant implanted therein, of the semiconductor substrate **10**.

[0083] An emitter region **12** of an N⁺ type is provided on the upper surface **21** of the semiconductor substrate **10**. The emitter region **12** in the present example is in contact with the metal layer **80** on the upper surface **21**. In the depth direction, a base region **14** of a P-type is provided between the emitter region **12** and the drift region **18**. An accumulation region **16** of the N⁺ type may be provided between the base region **14** and the drift region **18**. In another cross section of the active portion **120**, a contact region of the P⁺ type may be provided on the upper surface **21** of the semiconductor substrate **10**.

[0084] One or more trench portions **40** and one or more dummy trench portions **30** are provided in the semiconductor substrate **10** on a side of the upper surface **21**. The trench portion **40** may be a gate trench. Each trench portion extends from the upper surface **21** of the semiconductor substrate **10**, penetrating through the base region **14**, to reach the drift region **18**. In a region provided with at least one of the emitter region **12**, the contact region, or the accumulation region **16**, each trench portion also penetrates through the doping region, to reach the drift region **18**. Each trench portion penetrating through the doping region is not limited to manufacture performed by forming the doping region and then forming each trench portion. Each trench portion penetrating through the doping region also includes manufacture performed by forming each trench portion and then forming doping regions between respective trench portions. A region sandwiched between the respective trench portions within the semiconductor substrate **10** is referred to as a mesa portion **70**.

[0085] The trench portion **40** includes a trench provided on the upper surface **21** of the semiconductor substrate **10**, the dielectric film **42** and the conductive portion **44**. The dielectric film **42** is provided covering an inner wall of the trench. The dielectric film **42** may be formed by oxidizing or nitriding a semiconductor on the inner wall of the trench. The conductive portion **44** is provided farther inward than the dielectric film **42** within the trench. That is, the dielectric film **42** insulates the conductive portion **44** and the semiconductor substrate **10** from each other. The conductive portion **44** is made of a conductive material such as polysilicon.

[0086] The conductive portion **44** may be provided to be longer than the base region **14** in the depth direction. The conductive portion **44** is electrically connected to the gate runner **48**. In response to application of a predetermined gate voltage to the conductive portion **44**, a channel is formed by an electron inversion layer in a surface layer, which is in contact with the trench portion **40** and is a boundary surface between the base region **14** and the trench portion **40**, of the base region **14**.

[0087] The dummy trench portion **30** may have a same structure as that of the trench portion **40** in the cross section. The dummy trench portion **30** includes a dummy trench provided on the upper surface **21** of the semiconductor substrate **10**, a dummy dielectric film **32**, and a dummy conductive portion **34**. The dummy conductive portion **34** is electrically connected to the upper surface electrode **52** in another cross section. The dummy dielectric film **32** is provided covering an inner wall of the dummy trench. The dummy conductive portion **34** is provided within the dummy trench, and is provided farther inward than the dummy dielectric film **32**. The dummy dielectric film **32** insulates the dummy conductive portion **34** and the semiconductor substrate **10** from each other. The dummy conductive portion **34** may be made of a same material as that of the conductive portion **44**. For example, the dummy conductive portion **34** is made of a conductive material such as polysilicon. The dummy conductive portion **34** may have a same length as that of the conductive portion **44** in the depth direction.

[0088] The trench portion **40** and the dummy trench portion **30** in the present example are each covered with the first interlayer dielectric film **38-1** on the upper surface **21** of the semiconductor substrate **10**. It should be noted that bottom portions of the dummy trench portion **30** and the trench

portion **40** may each have a curved surface (be curved in the cross section) protruding downward. End portions, in the Y axis direction, of the trench portion **40** and the dummy trench portion **30** may be covered with the dividing well region **114** (see FIG. 2 and FIG. 4). It is to be noted that a gate provided in the semiconductor substrate **10** is not limited to a trench type. The semiconductor substrate **10** may be provided with a planar gate.

[0089] A buffer region **20** of the N+ type may be provided below the drift region **18**. A doping concentration in the buffer region **20** is higher than a doping concentration in the drift region **18**. The buffer region **20** may function as a field stopper layer which prevents a depletion layer expanding from a lower end of the base region **14** from reaching a collector region **22** of the P+ type.

[0090] The collector region **22** of the P+ type is provided below the buffer region **20**. An acceptor concentration in the collector region **22** is higher than an acceptor concentration in the base region **14**. The collector region **22** may include an acceptor which is the same as or different from an acceptor in the base region **14**. The acceptor in the collector region **22** is, for example, boron.

[0091] FIG. 6 illustrates one example of a top view of the semiconductor device **100**. FIG. 6 shows positions of the solder **35**, the lead frame **90**, and a protective film **94**.

[0092] The protective film **94** is provided above the upper surface **21** of the semiconductor substrate **10**. Protective films **94** are arranged so as to sandwich the active portion **120** of the semiconductor substrate **10** in a top view. The protective film **94** may enclose the active portion **120** in a top view. Dark hatching in FIG. 6 represents the protective film **94**. The protective film **94** separates the gate pad **50**, the current detection pad **72**, the anode pad **74**, and the cathode pad **76** from the active portion **120** in a top view. The protective film **94** may also separate portions of the upper surface electrode **52** which face respective control electrodes in the Y axis direction from portions thereof which do not face the respective control electrodes in the Y axis direction. The protective film **94** is, for example, polyimide.

[0093] In the active portion **120**, the solder **35** is provided above the upper surface electrode **52**. In a top view, a range provided with the solder **35** is defined by the protective film **94**. The solder **35** in the present example is formed over an entire area of the active portion **120** enclosed by the protective film **94**. In FIG. 6, a range where the solder **35** is formed is indicated by light hatching.

[0094] An end portion of the solder **35** in the direction in which the diode wiring **60** extends (the X axis direction) is referred to as an end portion position X_e. In other words, the solder **35** is provided extending to the end portion position X_e in the direction in which the diode wiring **60** extends. End portion positions X_e in the present example match end portions of the protective films **94** sandwiching the active portion **120**. It is to be noted that the end portion position X_e may not match an end portion of the protective film **94** enclosing the active portion **120**. The end portion position X_e may be located farther inward, at the semiconductor substrate **10**, than the end portion of the protective film **94** in a top view. In addition, if the end portion positions X_e differ from one position to another in a direction (the Y axis direction) perpendicular to the direction in which the diode wiring **60** extends, an end portion of the solder **35** at a position where the triple point G₁ (see FIG. 2) is formed may be referred to as the end portion position X_e. That is, the triple point G₁ is formed from a side of the temperature sensing diode **66** of the diode wiring **60** to the end portion position X_e. The lead frame **90** is provided above the solder **35**.

[0095] FIG. 7 illustrates one example of the semiconductor device **100** in a cross section D-D' shown in FIG. 6. The cross section D-D' is an XZ cross section crossing the protective film **94** from the diode wiring **60**. The solder **35** in the present example is also provided above the wiring protection film **92**. It is to be noted that the solder **35** may not be provided above the wiring protection film **92**.

[0096] The metal layer **80** is provided below the diode wiring **60** at the end portion position X_e in an extending direction in the present example (the X axis direction). Among the triple points described above, stress is particularly easily generated in a boundary portion where the solder **35** is

no longer formed. Therefore, a crack is easily formed starting from the triple point at the end portion position Xe. Providing the metal layer **80** below the end portion position Xe can increase strength of a part below the diode wiring **60** near a position where the crack is easily formed. If the diode wiring **60** and the end portion position Xe do not overlap with each other in the depth direction, the metal layer **80** may be provided below the diode wiring **60** at a position obtained by extending the end portion position Xe to a position where it overlaps with the diode wiring **60** in the Y axis direction.

[0097] The metal layer **80** may be provided extending to a position farther outward than the end portion position Xe of the solder **35** (a position on a negative side of an X axis). The metal layer **80** may be formed over a distance of at least 10 μm in the direction in which the diode wiring **60** extends, with the end portion position Xe as a center. The distance may be 20 μm , or may be 100 μm . The metal layer **80** may be provided continuously extending from the side of the temperature sensing diode **66** of the diode wiring **60** to the end portion position Xe.

[0098] In a top view, the metal layer **80** may be provided below an end portion of the protective film **94** overlapping with the diode wiring **60**. The end portion of the protective film **94** in the present example matches the end portion position Xe of the solder **35**. The end portion position Xe of the solder **35** is often defined by the end portion of the protective film **94**, so that providing the metal layer **80** below the end portion of the protective film **94** can increase the strength of the part below the diode wiring **60** near the position where the crack is easily formed.

[0099] The metal layer **80** may be provided extending to a position farther outward than the end portion of the protective film **94** (a position on the negative side of the X axis). The metal layer **80** may be formed over a distance of at least 10 μm in the direction in which the diode wiring **60** extends, with the end portion of the protective film **94** as a center. The distance may be 20 μm , or may be 100 μm . The metal layer **80** may be provided continuously extending from the side of the temperature sensing diode **66** of the diode wiring **60** to the end portion of the protective film **94**. The metal layer **80** may be provided below end portions of all the protective films **94** sandwiching the active portion **120**.

[0100] FIG. **8** illustrates a top view of one example of a semiconductor device **300** according to another embodiment of the present invention. The semiconductor device **300** in the present example is different from the semiconductor device **100** shown in FIG. **1B** in that the metal layer **80** is not formed below the temperature sensing diode **66**. Except for that, it is similar to the semiconductor device **100**. The metal layer **80** in the present example may also be provided extending to the active portion **120**.

[0101] FIG. **9** illustrates one example of the semiconductor device **300** in a cross section D-D' shown in FIG. **8**. The cross section D-D' is a YZ cross section perpendicular to the direction in which the diode wiring **60** extends (the X axis direction). The cross section D-D' is a cross section crossing the temperature sensing diode **66**.

[0102] As described above, in the semiconductor device **300**, the metal layer **80** is not formed below the temperature sensing diode **66**. The first interlayer dielectric film **38-1** and the gate runner **48** are formed below the temperature sensing diode **66** in the present example. Except for that, the semiconductor device **300** has a structure similar to that of the semiconductor device **100** shown in FIG. **4**. As described above, a crack is easily formed starting from the triple point at the end portion position Xe. The end portion position Xe of the solder **35** in the direction in which the diode wiring **60** extends (the X axis direction) in the present example does not overlap with the temperature sensing diode **66**. Therefore, there is relatively low possibility that the crack develops to below the temperature sensing diode **66**. Consequently, the metal layer **80** may not be provided below the temperature sensing diode **66**.

[0103] While the present invention has been described by way of the embodiments, the technical scope of the present invention is not limited to the above-described embodiments. It is apparent to persons skilled in the art that various alterations or improvements can be made to the above

described embodiments. It is also apparent from the description of the claims that embodiments added with such alterations or improvements can be included in the technical scope of the present invention.

Claims

1. A semiconductor device comprising a semiconductor substrate, wherein the semiconductor substrate has: a temperature sensing diode provided above an upper surface of the semiconductor substrate; a diode wiring provided above the upper surface of the semiconductor substrate and connected to the temperature sensing diode; a metal layer provided in at least part of a region between the diode wiring and the upper surface of the semiconductor substrate; a first interlayer dielectric film provided between the metal layer and the upper surface of the semiconductor substrate; and a second interlayer dielectric film provided between the diode wiring and the metal layer.
2. The semiconductor device according to claim 1, wherein the metal layer is also provided in at least part of a region between the temperature sensing diode and the upper surface of the semiconductor substrate.
3. The semiconductor device according to claim 1, wherein the metal layer contains a metal harder than aluminum.
4. The semiconductor device according to claim 3, wherein the metal layer contains at least one of tungsten, titanium, or tantalum.
5. The semiconductor device according to claim 1, further comprising an upper surface electrode provided above the upper surface of the semiconductor substrate and in contact with part of the upper surface of the semiconductor substrate, wherein the upper surface electrode is connected to the metal layer.
6. The semiconductor device according to claim 5, wherein the upper surface electrode does not overlap with the diode wiring in a top view.
7. The semiconductor device according to claim 6, wherein the metal layer contains a harder metal than the upper surface electrode.
8. The semiconductor device according to claim 6, wherein the metal layer is in contact with the upper surface of the semiconductor substrate in a region which does not overlap with the diode wiring and the temperature sensing diode.
9. The semiconductor device according to claim 6, wherein a side surface of the second interlayer dielectric film is in contact with the upper surface electrode.
10. The semiconductor device according to claim 6, wherein the upper surface electrode includes a first portion and a second portion sandwiching the diode wiring in a top view, and the metal layer is provided extending from the first portion, passing below the diode wiring, to the second portion.
11. The semiconductor device according to claim 6, wherein the semiconductor substrate further has: a trench portion provided extending from the upper surface to an inside of the semiconductor substrate and having a conductive portion provided therein; and a well region provided extending from the upper surface of the semiconductor substrate to a depth greater than that of a lower end of the trench portion, the diode wiring overlaps with the well region in a top view, and the first interlayer dielectric film is provided with a contact hole which electrically connects the metal layer and the well region to each other.
12. The semiconductor device according to claim 11, wherein the semiconductor substrate further has active portions each provided with a semiconductor element, and the well region is sandwiched between the active portions in a top view.
13. The semiconductor device according to claim 11, wherein in a cross section perpendicular to a direction in which the diode wiring extends, a width of the second interlayer dielectric film in a horizontal direction parallel to the upper surface of the semiconductor substrate is smaller than a

width of the well region.

14. The semiconductor device according to claim 11, wherein the first interlayer dielectric film is provided with two contact holes including the contact hole, the semiconductor substrate further has a gate runner provided above the upper surface of the semiconductor substrate, and the gate runner is provided between the two contact holes in a top view.

15. The semiconductor device according to claim 1, further comprising: an upper surface electrode provided above the upper surface of the semiconductor substrate and in contact with part of the upper surface of the semiconductor substrate; and solder at least part of which is provided above the upper surface electrode, wherein in a direction in which the diode wiring extends, the solder is provided extending to an end portion position, and the metal layer is provided below the diode wiring at the end portion position in the direction in which the diode wiring extends.

16. The semiconductor device according to claim 1, further comprising protective films provided above the semiconductor substrate and arranged so as to sandwich an active portion of the semiconductor substrate in a top view, wherein in a top view, the metal layer is provided below end portions of the protective films overlapping with the diode wiring.

17. The semiconductor device according to claim 2, further comprising an upper surface electrode provided above the upper surface of the semiconductor substrate and in contact with part of the upper surface of the semiconductor substrate, wherein the upper surface electrode is connected to the metal layer.

18. The semiconductor device according to claim 3, further comprising an upper surface electrode provided above the upper surface of the semiconductor substrate and in contact with part of the upper surface of the semiconductor substrate, wherein the upper surface electrode is connected to the metal layer.

19. The semiconductor device according to claim 4, further comprising an upper surface electrode provided above the upper surface of the semiconductor substrate and in contact with part of the upper surface of the semiconductor substrate, wherein the upper surface electrode is connected to the metal layer.

20. The semiconductor device according to claim 2, further comprising: an upper surface electrode provided above the upper surface of the semiconductor substrate and in contact with part of the upper surface of the semiconductor substrate; and solder at least part of which is provided above the upper surface electrode, wherein in a direction in which the diode wiring extends, the solder is provided extending to an end portion position, and the metal layer is provided below the diode wiring at the end portion position in the direction in which the diode wiring extends.
